

PROFILE

I explain AI systems from having worked through the pipeline myself: collecting training data, synthesising verifiable training environments, RL post-training, product integration, guardrails, demos, and public technical content. In open source, I worked on terminal-agent RL training pipelines reaching SOTA at 8B scale; outside work, I used seven hackathons in four weeks to learn AI-assisted product building across frontend, backend, databases, demos, and pitches.

EXPERIENCE

AI Data Scientist, Tesco

Winter 2024 – Present

Production LLM Agent and Applied AI Systems

- Collaborated with OpenAI-acquired Tomoro on a large-scale LLM agentic system that turns customer conversations and meal ideas into recipe plans, purchasable Tesco products, and shopping baskets; shipped a **10-week beta** for **280k internal users**
- Built the agent improvement loop: Langfuse traces over plans, tool calls and state transitions; tiered evaluators for tool use, execution, answer quality and end-to-end outcomes; online annotation to calibrate LLM-as-a-Judge scoring and release gates
- Scaled retail-specific red-team testing with Promptfoo; converted failure modes into layered guardrails, output retry policies, prompt updates, and agent architecture and harness improvements

Applied ML, Retrieval, and Data Systems

- Led a recipe-to-basket recommendation PoC, adapting LightRAG into a recipe–ingredient–SKU graph and dual retrieval system grounded in live product data
- Productionised PySpark ETL pipelines and product-substitution models across **3k+ stores**, delivering **£5M+ annualised savings**; used LLM semantic similarity to increase substitute coverage by **32.51%**

Off-Cycle Intern, Global Banking & Markets Division, Goldman Sachs

Autumn 2024

- Built Slang pipelines converting unstructured financial streams into structured daily market insights, replacing manual workflows

RESEARCH & OPEN SOURCE

SETA: Scaling Environments for Terminal Agents, Second Author & Contributor, CAMEL-AI

Winter 2025 – Present

- Built **SETA-Env**, the largest open-source verifiable terminal RL environment dataset to date with **4.5k+** Docker-isolated environments for terminal agents
- Designed **SETA-Synth** and **SETA-Evol**: data synthesis pipelines that expand task diversity, evolve task difficulty to the model capability frontier, and filter invalid environments with a Trajectory Judge Agent
- Integrated CAMEL agents with AReaL/MILES, distributed rollout workers, remote env-service nodes and SGLang inference; Qwen3-8B reached **12.0%** on Terminal-Bench 2.0 (3.3x), achieving **SOTA at 8B scale** and beating **GPT-5 nano**

Blockchain Coder: Multi-Agent Code Generation Framework, UCL

Summer 2024

- Designed a multi-agent framework (Planner, Optimizer, Debugger) to improve held-out generalisation beyond memorisation
- Used unit-test-driven interactive refinement to double pass@1 to 6.90% and improve private-test pass rate by **5.7x** to 38.08%

EDUCATION

University College London | MSc Machine Learning | First Class

2023 – 2024

- Multi-agent AI (85.5%), Information Retrieval and Data Mining (87.5%), Bayesian Deep Learning (83.5%)

Imperial College London | BSc Mathematics | First Class (80.3%)

2020 – 2023

- Methods for Data Science (94.4%), Mathematical Foundations of Machine Learning (90%)

TECHNICAL SKILLS

Languages	Python, SQL, R, Julia; Mandarin (Native), Cantonese (Native), English (Fluent)
Developer Content	Technical tutorials, demo building, AI content creation, GitHub projects, developer education, product walkthroughs
AI Systems	Production LLM agents, RAG/LightRAG, LangChain/Langfuse, Promptfoo/DSPy, PyTorch, HF, AutoGen

ACHIEVEMENTS

- 5x hackathon winner with £8k+ in prizes; projects include Sign2Speak, SquareDeal, and CartPilot, an AI shopping assistant combining purchase prediction, RL-based recommendations, and Web3-wallet checkout automation for Walmart
- AI content creator: open-sourced MarketVision, an AI stock tool, and gained 1k+ followers from one tutorial video